Berk Iskender

Boston, MA, US | Personal email | +12177213104 | Personal Website | LinkedIn | GitHub | Scholar

Summary

Ph.D. in Electrical and Computer Engineering with 8 years of research experience in computational imaging, computer vision, machine learning, signal processing, and inverse problems, resulting in multiple first-author publications in peer-reviewed journals and conferences.

Education

University of Illinois Urbana-Champaign (UIUC), IL, US, GPA: 3.94 / 4.00,
Ph.D. in Electrical and Computer Engineering **Advisor**: Prof. Yoram Bresler 2020 – 2024

University of Illinois Urbana-Champaign (UIUC), IL, US, GPA: 3.94 / 4.00,
M.S. in Electrical and Computer Engineering **Advisor**: Prof. Yoram Bresler 2018 – 2020

Middle East Technical University (METU), Ankara, TR, GPA: 4.00 / 4.00,
B.S. in Electrical Engineering, *Valedictorian* 2014 – 2018

Experience

Senior Machine Learning Researcher/Engineer July 2024 – Present
Analog Garage (research incubator) at Analog Devices, Inc. (ADI) – Boston, MA, US

- Currently working on a research project focusing on an inverse problem in computational RF imaging.
- Conducted research to develop an end-to-end pipeline for 3D scene reconstruction and semantic segmentation of RGB-D data with object tracking. Leveraged the foundational Segment Anything Model 2 (SAM2) for segmentation and tracking, where the project also combined state-of-the-art semantic segmentation algorithms, and semantics-guided point cloud fusion. (Manuscript submitted to IEEE IROS 2025)
- Implemented navigation capabilities for an omnidirectional robot using the ROS2 Nav2 stack.

PhD SWE (Machine Learning) Internship & Student Researcher May 2022 - Oct 2022
Google – Mountain View, CA, US

- Research on improving self-supervised dense contrastive learning of uncurated data using transformers, different dense comparison techniques, and reconstruction decoders (published in NeurIPS 2022 Self-Supervised Learning - Theory & Practice Workshop).

PhD SWE (Machine Learning) Internship May 2021 - Aug 2021
Google – Mountain View, CA, US

- Implemented & compared various visual-semantic image embedding models. Deployed a supervised contrastive learning method to replace the attribute-based embedding to assist graph-hierarchical clustering at Google Geo.

Graduate Research Internship May 2020 - Aug 2020
Michigan State University (MSU), MI, US

- Research on block-matching algorithms with learned sparsifying transforms for image denoising (published in IEEE ICIP 2021) and dynamic estimation methods. **Advisor**: Prof. Saiprasad Ravishankar

Graduate Research Internship July 2019 - Aug 2019
Los Alamos National Laboratory (LANL), NM, US

- Worked on the deep learning-based methods for ill-posed single-view tomographic reconstruction.

Undergraduate Research Internship July 2017 - Aug 2017
ASELSAN Research Dept., Ankara, TR

- Worked on time/frequency domain passive acoustic mapping, sparsity-based microbubble detection with constrained optimization for ultrasound.

Undergraduate Internship June 2016 - July 2016
KAREL Research & Development Center, Ankara, TR

- Performed image processing tasks for vehicles on ARM NXP iMX6 by cross-compilation of OpenCV libraries.

Professional Services

Teaching Graduate teaching assistant of Digital Signal Processing course for Fall 2019, Spring 2020 and Fall 2020.

Reviewer IEEE Transactions in Medical Imaging, IEEE Transactions in Computational Imaging, IEEE ICASSP, IEEE/CVF CVPR, The Journal of Supercomputing.

Interests

Interested in machine learning, computer vision, signal processing, computational imaging & inverse problems.

Qualifications

Programming	Core: Python, Matlab, LaTeX Used for various tasks: C/C++, SQL, HTML, ARM Assembly
Libraries/Frameworks	Pytorch, TensorFlow, ROS2, ROS2 Nav2 (Navigation stack), Pandas, Scikit-learn
Application/Software	Github, OpenCV, LabView, GEANT4 Used for coursework: Altera Quartus, LTSpice, CAD
Languages	English (Proficient), Turkish (Native), Spanish (Pre-intermediate)

Publications

- **B. Iskender**, et. al., "RSR-NF: Neural Field Regularization by Static Restoration Priors for Dynamic Imaging", IEEE MLSP (Oral Presentation) & arXiv:2503.10015, 2025.
- **B. Iskender**, M. Klasky, Y. Bresler, "RED-PSM: Regularization by Denoising of Factorized Low Rank Models for Dynamic Imaging", IEEE Transactions on Computational Imaging, 2024.
- Z. Zheng, L. Mentzer, **B. Iskender**, M. Price, C. Prendergast, A. Cloitre, "Semantic Segmentation and Scene Reconstruction of RGB-D Image Frames: An End-to-End Modular Pipeline for Robotic Applications", arXiv:2410.17988, 2024.
- **B. Iskender**, M. Klasky, Y. Bresler, "Dynamic Tomography Reconstruction via Low-Rank Modeling with a RED Spatial Prior", IEEE CAMSAP 2023.
- **B. Iskender**, M. Klasky, Y. Bresler, "RED-PSM: Regularization by Denoising of Partially Separable Models for Dynamic Imaging", IEEE/CVF International Conference on Computer Vision (ICCV) 2023.
- **B. Iskender**, M. Klasky, B. Patterson, Y. Bresler, "Factorized Projection-domain Spatio-temporal Regularization for Dynamic Tomography", IEEE ICASSP 2023.
- **B. Iskender**, et. al., "Improving Dense Contrastive Learning with Dense Negative Pairs", NeurIPS 2022: Self-Supervised Learning: Theory and Practice Workshop.
- **B. Iskender**, M. Klasky, Y. Bresler, "Dynamic Tomography Reconstruction by Projection-Domain Separable Modeling", IEEE IVMSP 2022 & arXiv:2204.09935.
- **B. Iskender**, Y. Bresler, "Scatter Correction in X-ray CT by Physics-Inspired Deep Learning", IEEE Transactions on Computational Imaging, 2022.
- S. Liang, **B. Iskender**, B. Wen, S. Ravishankar, "Learned feature-domain block matching for image restoration", IEEE Intl. Conf. on Image Processing (ICIP) 2021.
- **B. Iskender**, Y. Bresler, "A physics-motivated DNN for X-ray CT scatter correction", IEEE 17th Intl. Symposium on Biomedical Imaging, (ISBI) 2020.
- **B. Iskender**, Y. Bresler, "X-ray CT scatter correction by a physics-motivated DNN with opposite view processing", CT Meeting 2020.
- **B. Iskender**, S.F. Oktem, "Image restoration for sparse aperture optical systems", IEEE 26th Signal Proc. and Comm. Applications Conference (SIU), 2018.

Theses

- Ph.D.: Iskender, B. (2024). Spatio-temporal tomographic imaging (University of Illinois at Urbana-Champaign).
- M.S.: Iskender, B. (2020). X-ray CT scatter correction by a physics-motivated deep neural network (University of Illinois at Urbana-Champaign).

Selected Courses & Projects

- **Computer Vision**: Project on SoTA algorithms comparison for agricultural image segmentation, Coursework: image & shape retrieval, image registration, optical flow calculations
- **Machine Learning**: Project on supervised image super-resolution and denoising on low-dose X-ray images
- **Machine Learning for Signal Processing**: Project on developing a super-resolution objective maximizing Fourier shell correlation for a GAN-based model
- **Generative AI Models**: Project on application of deep prior generative models for video reconstruction to dynamic tomography
- **Digital Imaging**: Project on CNN-based projected gradient descent for consistent X-ray CT scatter correction
- **Vector Space Signal Processing**: Project on reducing spatially varying out-of-focus blur
- **Computational Inference and Learning**: Project on LASSO problem analysis and comparison of various analytic/learning-based algorithms
- **Senior Design Project**: Implemented computer vision tasks (shape detection, motion control, and decision) of a basketball-playing robot on Raspberry Pi 3
- **Digital Signal Processing II**: Project on solving ill-posed inverse problems using ML/classical models
- **Probability & Random Var.**: Project on maximum likelihood estimation from observations in moving object detection
- **Communications**: Project on hypothesis testing for amplitude or frequency-modulated signals
- **Random Processes**
- **Convex Optimization**

Achievements and Awards

- Ranked 1st in the Electrical Engineering department, and Valedictorian at Middle East Technical University in 2018.
- Dean's High Honor List, Middle East Technical University (All semesters)
- Middle East Technical University Electrical Engineering Bulent Kerim Altay Award (6 times) (Highest academic performance award for the related semester given by the EE department)
- Top 100th student scholarship in the national university entrance exam given by the Turkish Ministry of Education
- IEEE Signal Processing Society Student Travel Award & Student Best Paper Award finalist of CAMSAP 2023
- IEEE Student Travel Award for ISBI 2020